Color Factor Evaluation: Every Step

From $\dfrac{1}{24}\sum_{a,b}\mathrm{Tr}[\lambda^a\lambda^b\lambda^b\lambda^a]$ to $\dfrac{32}{9}$

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1 Starting Point

We showed that after squaring, using Hermiticity, summing over quark colors (i, j), and averaging, the color factor is:

$$\mathcal{C} = \frac{1}{24}\sum_{a=1}^{8}\sum_{b=1}^{8}\mathrm{Tr}\Big[\lambda^a\lambda^b\lambda^b\lambda^a\Big] \tag{1}$$

We need to evaluate this. The strategy is: sum over b first (the inner sum), then sum over a.

2 Why We Can Sum Over b First

The trace is:

$$\mathrm{Tr}\Big[\lambda^a\lambda^b\lambda^b\lambda^a\Big] = \mathrm{Tr}\Big[\lambda^a\,\left(\lambda^b\lambda^b\right)\,\lambda^a\Big] \tag{2}$$

The sum $\sum_b$ only acts on the $\lambda^b\lambda^b$ part. The λ^a matrices don't depend on b, so they just sit there while we sum the middle:

$$\sum_{b=1}^{8}\mathrm{Tr}\Big[\lambda^a\lambda^b\lambda^b\lambda^a\Big] = \mathrm{Tr}\left[\lambda^a\quad\underbrace{\left(\sum_{b=1}^{8}\lambda^b\lambda^b\right)}_{\text{evaluate this object}}\quad\lambda^a\right] \tag{3}$$

This works because the trace and the sum are both linear operations. The sum over b produces a single 3×3 matrix, which then sits between the two λ^a's inside the trace.

3 Evaluating $\sum_{b=1}^{8}\lambda^b\lambda^b$

3.1 What This Object Is

$\sum_{b=1}^{8}\lambda^b\lambda^b$ is a 3×3 matrix. It is the sum of 8 matrix products. Let's compute it element by element.

3.2 The (m, n) Element

$$\left(\sum_{b=1}^{8}\lambda^b\lambda^b\right)_{mn} = \sum_{b=1}^{8}\sum_{k=1}^{3}(\lambda^b)_{mk}\,(\lambda^b)_{kn} \tag{4}$$

This is the (m, n) element of the matrix product $\lambda^b\lambda^b$, summed over all 8 gluon types.

3.3 Connection to the Casimir Operator

The SU(3) generators in the fundamental representation are $T^b = \lambda^b/2$. The quadratic Casimir operator is defined as:

$$\hat{C}_2 \equiv \sum_{b=1}^{8} T^b T^b = \sum_{b=1}^{8} \frac{\lambda^b}{2} \cdot \frac{\lambda^b}{2} = \frac{1}{4} \sum_{b=1}^{8} \lambda^b \lambda^b \tag{5}$$

For any simple Lie group, this operator is proportional to the identity matrix:

$$\hat{C}_2 = C_F \, \mathbb{I} \tag{6}$$

where C_F is a number called the Casimir invariant. For SU(N) in the fundamental representation:

$$C_F = \frac{N^2 - 1}{2N} \tag{7}$$

For SU(3): $C_F = \frac{9-1}{6} = \frac{8}{6} = \frac{4}{3}$.
Therefore:

$$\frac{1}{4} \sum_{b=1}^{8} \lambda^b \lambda^b = \frac{4}{3} \mathbb{I}_3 \tag{8}$$

Multiply both sides by 4:

$$\boxed{\sum_{b=1}^{8} \lambda^b \lambda^b = \frac{16}{3} \mathbb{I}_3 = \frac{16}{3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}} \tag{9}$$

Why Is It Proportional to the Identity?

By Schur's lemma: if an operator commutes with every generator of the group, and the representation is irreducible, then it must be proportional to the identity.
$\hat{C}_2 = \sum_b T^b T^b$ commutes with every T^a (you can verify $[\hat{C}_2, T^a] = 0$). The fundamental representation of SU(3) is irreducible. Therefore $\hat{C}_2 \propto \mathbb{I}$.

3.4 Explicit Verification: Every Element of $\sum_b \lambda^b \lambda^b$

Let's verify (9) by computing every element. Using (4):

$$\left(\sum_b \lambda^b \lambda^b \right)_{mn} = \sum_{b=1}^{8} \sum_{k=1}^{3} (\lambda^b)_{mk} \, (\lambda^b)_{kn} \tag{10}$$

3.4.1 Diagonal element $(1,1)$: should give $16/3$

Need $\sum_b \sum_k (\lambda^b)_{1k} (\lambda^b)_{k1}$. Since λ^b is Hermitian, $(\lambda^b)_{k1} = [(\lambda^b)_{1k}]^*$, so this is $\sum_b \sum_k |(\lambda^b)_{1k}|^2$.
Go through each λ^b, listing the nonzero entries in row 1:

λ^b	$k=1$	$k=2$	$k=3$
λ^1	$(\lambda^1)_{11}=0$	$(\lambda^1)_{12}=1$	$(\lambda^1)_{13}=0$
λ^2	$(\lambda^2)_{11}=0$	$(\lambda^2)_{12}=-i$	$(\lambda^2)_{13}=0$
λ^3	$(\lambda^3)_{11}=1$	$(\lambda^3)_{12}=0$	$(\lambda^3)_{13}=0$
λ^4	$(\lambda^4)_{11}=0$	$(\lambda^4)_{12}=0$	$(\lambda^4)_{13}=1$
λ^5	$(\lambda^5)_{11}=0$	$(\lambda^5)_{12}=0$	$(\lambda^5)_{13}=-i$
λ^6	0	0	0
λ^7	0	0	0
λ^8	$(\lambda^8)_{11}=\frac{1}{\sqrt{3}}$	0	0

Sum of $|(\lambda^b)_{1k}|^2$ over all b and k:

$$\begin{aligned}
&= |1|^2 + |-i|^2 + |1|^2 + |1|^2 + |-i|^2 + 0 + 0 + \left|\tfrac{1}{\sqrt{3}}\right|^2 \\
&= 1+1+1+1+1+0+0+\frac{1}{3} \\
&= 5+\frac{1}{3} = \frac{16}{3} \quad \checkmark
\end{aligned} \tag{11}$$

3.4.2 Diagonal element $(2,2)$: should also give $16/3$

Need $\sum_b \sum_k |(\lambda^b)_{2k}|^2$. Nonzero entries in row 2:

λ^b	$k=1$	$k=2$	$k=3$
λ^1	$(\lambda^1)_{21}=1$	0	0
λ^2	$(\lambda^2)_{21}=i$	0	0
λ^3	0	$(\lambda^3)_{22}=-1$	0
λ^4	0	0	0
λ^5	0	0	0
λ^6	0	0	$(\lambda^6)_{23}=1$
λ^7	0	0	$(\lambda^7)_{23}=-i$
λ^8	0	$(\lambda^8)_{22}=\frac{1}{\sqrt{3}}$	0

$$\begin{aligned}
&= |1|^2 + |i|^2 + |-1|^2 + 0 + 0 + |1|^2 + |-i|^2 + \left|\tfrac{1}{\sqrt{3}}\right|^2 \\
&= 1+1+1+1+1+\frac{1}{3} = \frac{16}{3} \quad \checkmark
\end{aligned} \tag{12}$$

3.4.3 Diagonal element $(3,3)$: should also give $16/3$

Need $\sum_b \sum_k |(\lambda^b)_{3k}|^2$. Nonzero entries in row 3:

λ^b	$k=1$	$k=2$	$k=3$
λ^4	$(\lambda^4)_{31}=1$	0	0
λ^5	$(\lambda^5)_{31}=i$	0	0
λ^6	0	$(\lambda^6)_{32}=1$	0
λ^7	0	$(\lambda^7)_{32}=i$	0
λ^8	0	0	$(\lambda^8)_{33}=\frac{-2}{\sqrt{3}}$

(All other λ^b have row 3 entirely zero.)

$$= |1|^2 + |i|^2 + |1|^2 + |i|^2 + \left|\tfrac{-2}{\sqrt{3}}\right|^2$$
$$= 1+1+1+1+\frac{4}{3} = 4+\frac{4}{3} = \frac{16}{3} \quad \checkmark \tag{13}$$

3.4.4 Off-diagonal element $(1,2)$: should give 0

Need $\sum_b \sum_k (\lambda^b)_{1k}(\lambda^b)_{k2}$.

The only λ^b matrices with nonzero entries in *both* row 1 and column 2 are:

λ^b	$(\lambda^b)_{1k}$	$(\lambda^b)_{k2}$ for same k
λ^1	$(\lambda^1)_{12}=1$	need $(\lambda^1)_{22}=0$. Also $(\lambda^1)_{11}=0$, $(\lambda^1)_{12}=1$ with $(\lambda^1)_{22}=0$
λ^2	$(\lambda^2)_{12}=-i$	$(\lambda^2)_{22}=0$
λ^3	$(\lambda^3)_{11}=1$	$(\lambda^3)_{12}=0$
λ^8	$(\lambda^8)_{11}=\frac{1}{\sqrt{3}}$	$(\lambda^8)_{12}=0$

Let me be systematic. For each b, compute $\sum_k (\lambda^b)_{1k}(\lambda^b)_{k2}$:
λ^1: $(\lambda^1)_{11}(\lambda^1)_{12} + (\lambda^1)_{12}(\lambda^1)_{22} + (\lambda^1)_{13}(\lambda^1)_{32} = 0\cdot 1 + 1\cdot 0 + 0\cdot 0 = 0$
λ^2: $0\cdot(-i) + (-i)\cdot 0 + 0 = 0$
λ^3: $1\cdot 0 + 0\cdot(-1) + 0 = 0$
λ^4: $0+0+1\cdot 0 = 0$
λ^5: $0+0+(-i)\cdot 0 = 0$
λ^6: all row-1 entries are 0, so contribution is 0
λ^7: all row-1 entries are 0, so contribution is 0
λ^8: $\frac{1}{\sqrt{3}}\cdot 0 + 0\cdot\frac{1}{\sqrt{3}} + 0 = 0$
Total: 0 ✓

Verified

$$\sum_{b=1}^{8} \lambda^b \lambda^b = \frac{16}{3}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{14}$$

Diagonal elements all give 16/3. Off-diagonal elements all give 0. (We checked $(1,2)$; the other off-diagonals work the same way by the SU(3) structure.)

4 Substituting Back Into the Trace

From (3), after summing over b:

$$\sum_{b=1}^{8} \mathrm{Tr}\Big[\lambda^a \lambda^b \lambda^b \lambda^a\Big] = \mathrm{Tr}\Big[\lambda^a \cdot \frac{16}{3}\,\mathbb{I}_3 \cdot \lambda^a\Big] \tag{15}$$

Now the $\frac{16}{3}\,\mathbb{I}_3$ is a scalar (16/3) times the identity matrix. Multiplying any matrix by a scalar times the identity just gives that scalar times the matrix:

$$\lambda^a \cdot \frac{16}{3}\,\mathbb{I}_3 = \frac{16}{3}\,\lambda^a \tag{16}$$

So:

$$\mathrm{Tr}\Big[\lambda^a \cdot \frac{16}{3}\,\mathbb{I}_3 \cdot \lambda^a\Big] = \mathrm{Tr}\Big[\frac{16}{3}\,\lambda^a \cdot \lambda^a\Big] = \frac{16}{3}\,\mathrm{Tr}[\lambda^a \lambda^a] \tag{17}$$

where in the last step we pulled the scalar 16/3 out of the trace (the trace is linear: $\mathrm{Tr}(cA) = c\,\mathrm{Tr}(A)$).

Substituting into (1):

$$\mathcal{C} = \frac{1}{24}\sum_{a=1}^{8} \frac{16}{3}\,\mathrm{Tr}(\lambda^a \lambda^a) = \frac{16}{3 \times 24}\sum_{a=1}^{8} \mathrm{Tr}(\lambda^a \lambda^a) = \frac{16}{72}\sum_{a=1}^{8} \mathrm{Tr}(\lambda^a \lambda^a) \tag{18}$$

(pulled the constant 16/3 out of the sum over a, since it doesn't depend on a).

5 Evaluating $\mathrm{Tr}(\lambda^a \lambda^a)$ for Each a

The trace identity $\mathrm{Tr}(\lambda^a \lambda^b) = 2\delta^{ab}$ (Griffiths Eq. 9.26) tells us:

$$\mathrm{Tr}(\lambda^a \lambda^a) = 2\delta^{aa} = 2 \times 1 = 2 \qquad \text{(no sum on } a\text{)} \tag{19}$$

But let's verify this explicitly for a few cases.

5.1 $a = 1$: $\mathrm{Tr}(\lambda^1 \lambda^1)$

$$\lambda^1 \lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{20}$$

$$\mathrm{Tr}(\lambda^1 \lambda^1) = 1 + 1 + 0 = 2 \quad \checkmark \tag{21}$$

5.2 $a = 3$: $\mathrm{Tr}(\lambda^3 \lambda^3)$

$$\lambda^3 \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{22}$$

$$\mathrm{Tr}(\lambda^3 \lambda^3) = 1 + 1 + 0 = 2 \quad \checkmark \tag{23}$$

5.3 $a = 8$: $\mathbf{Tr}(\lambda^8\lambda^8)$

$$\lambda^8\lambda^8 = \frac{1}{3}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-2\end{pmatrix}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-2\end{pmatrix} = \frac{1}{3}\begin{pmatrix}1&0&0\\0&1&0\\0&0&4\end{pmatrix} \tag{24}$$

$$\mathrm{Tr}(\lambda^8\lambda^8) = \frac{1}{3}(1+1+4) = \frac{6}{3} = 2 \quad \checkmark \tag{25}$$

5.4 $a = 2$: $\mathbf{Tr}(\lambda^2\lambda^2)$

$$\lambda^2\lambda^2 = \begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix} = \begin{pmatrix}(-i)(i)&0&0\\0&(i)(-i)&0\\0&0&0\end{pmatrix} = \begin{pmatrix}1&0&0\\0&1&0\\0&0&0\end{pmatrix} \tag{26}$$

$$\mathrm{Tr}(\lambda^2\lambda^2) = 1+1+0 = 2 \quad \checkmark \tag{27}$$

Trace Identity Verified

$$\mathrm{Tr}(\lambda^a\lambda^a) = 2 \qquad \text{for every } a = 1, 2, \ldots, 8 \tag{28}$$

This is the same value for every a because the λ matrices are normalized to satisfy $\mathrm{Tr}(\lambda^a\lambda^b) = 2\delta^{ab}$.

6 The Sum Over a

Since $\mathrm{Tr}(\lambda^a\lambda^a) = 2$ for *every* a, the sum is trivial:

$$\sum_{a=1}^{8}\mathrm{Tr}(\lambda^a\lambda^a) = \underbrace{2+2+2+2+2+2+2+2}_{8\text{ terms}} = 2\times 8 = 16 \tag{29}$$

7 Final Arithmetic

Substituting (29) into (18):

$$\mathcal{C} = \frac{16}{72}\times 16 \tag{30}$$

$$= \frac{16\times 16}{72} \tag{31}$$

$$= \frac{256}{72} \tag{32}$$

Simplify: $\gcd(256, 72) = 8$, so:

$$\frac{256}{72} = \frac{256\boldsymbol{\nabla}\cdot 8}{72\boldsymbol{\nabla}\cdot 8} = \frac{32}{9} \tag{33}$$

Color Factor

$$\boxed{\mathcal{C} = \frac{32}{9}} \tag{34}$$

Breakdown of where each number came from:

$$\mathcal{C} = \underbrace{\frac{1}{24}}_{\frac{1}{3}\times\frac{1}{8}\text{ (avg)}} \times \underbrace{\frac{16}{3}}_{\sum_b \lambda^b\lambda^b = \frac{16}{3}\mathbb{I}\text{ (Casimir)}} \times \underbrace{16}_{\sum_a \mathrm{Tr}(\lambda^a\lambda^a)\text{ (trace id.)}}$$

$$= \frac{1}{24} \times \frac{16}{3} \times 16 = \frac{1}{24} \times \frac{256}{3} = \frac{256}{72} = \frac{32}{9}$$